AZURE ENTERPRISE INFRASTRUCTURE LAB

06 — Identity & Security

Managed Identity + Azure RBAC + Key Vault

Evidence-based implementation documentation with attached Azure Portal screenshots

1. Objective

Demonstrate Azure identity and security fundamentals by reviewing RBAC scope, enabling a system-assigned managed identity on the Windows VM, creating an Azure Key Vault with Azure RBAC, assigning the VM the Key Vault Secrets User role, and troubleshooting a data-plane permission error while creating a test secret.

2. Environment

Resource / Setting	Observed value
Subscription	My Test Azure subscription 1
Resource Group	rg-azure-enterprise_MM
VM	vm-web-MM
VM OS	Windows Server 2025 Datacenter
VNet	vnet-azure-enterprise-MM
Subnet	WebSubnet_MM
VM Private IP	10.0.1.4
VM Public IP	20.244.97.131
Key Vault	kv-azure-enterprise-mm
Key Vault region	Central India
Key Vault pricing tier	Standard
Key Vault permission model	Azure role-based access control

3. Security Architecture

- User / administrator → Azure RBAC at subscription/resource scopes
- vm-web-MM → System-assigned Managed Identity → Microsoft Entra ID
- Managed Identity → Key Vault Secrets User → kv-azure-enterprise-mm
- Key Vault → Secrets → DemoSecret (planned test secret; creation currently blocked by the user's own data-plane permissions)

4. RBAC Scope Review

The subscription IAM screenshot shows the signed-in account has an active Owner role at subscription scope. The VM IAM screenshot shows that Owner is inherited from the subscription.

No additional Owner or Contributor assignment is required for the administrator. This is useful evidence for explaining RBAC scope inheritance and least-privilege design.

5. Managed Identity Configuration

A system-assigned managed identity was enabled on vm-web-MM. The identity is tied to the lifecycle of the VM and can be granted Azure RBAC permissions without embedding credentials in code or configuration.

Verified state: System assigned = On.

6. Key Vault Creation

- Key Vault name: kv-azure-enterprise-mm
- Resource group: rg-azure-enterprise_MM
- Region: Central India
- Pricing tier: Standard
- Soft delete: Enabled
- Retention: 90 days
- Purge protection: Disabled
- Permission model: Azure RBAC
- Networking: Public endpoint / all networks for the initial lab

7. Key Vault RBAC Assignment

The VM managed identity was assigned the built-in role:

Key Vault Secrets User

Scope: kv-azure-enterprise-mm. Member: vm-web-MM (Virtual machine managed identity).

This role is intended for reading secret values; it is not sufficient for the signed-in administrator to create or manage secrets.

8. Troubleshooting: Secret Creation Forbidden

A test secret named DemoSecret was prepared with a dummy value, but creation failed with a Forbidden/RBAC error.

Observed issue: The portal reported that the operation was not allowed by RBAC and advised allowing several minutes for recently changed role assignments to become effective. The caller was not authorized to perform the secret set operation.

This behavior is consistent with a separation between Azure resource-plane administration and Key Vault data-plane permissions. Having Owner at subscription/resource scope does not by itself grant the required Key Vault secret data action under the Azure RBAC permission model.

9. Correct Remediation

To allow the administrator to create/manage test secrets while preserving the VM's least-privilege read access:

1. Key Vault → Access control (IAM) → Add role assignment

2. Role: Key Vault Secrets Officer
3. Assign access to: User, group, or service principal
4. Member: the administrator's Azure account
5. Review + assign
6. Allow several minutes for RBAC propagation, then refresh/retry

Do not remove the VM's Key Vault Secrets User role. Keep it so the VM can later retrieve secrets without write permissions.

10. Planned Test Secret

- Name: DemoSecret
- Value: AzureLab-Test-2026
- Enabled: Yes

Use this dummy value only. Never use a production password, API token, connection string, or other real credential in a learning secret or public GitHub repository.

11. Validation Plan

7. Create DemoSecret successfully from the administrator account after the Secrets Officer role propagates.
8. Keep vm-web-MM as Key Vault Secrets User.
9. From vm-web-MM, authenticate with its managed identity and perform a read-only secret retrieval test.
10. Capture successful role-assignment, secret listing, and read-validation screenshots.
11. For a production-style design, replace the public Key Vault endpoint with private connectivity and restrict network access.

12. Interview Skills Demonstrated

- Azure RBAC and scope inheritance
- Least privilege
- System-assigned managed identity
- Microsoft Entra ID integration
- Azure Key Vault
- Key Vault RBAC
- Secrets management
- Data-plane vs control-plane permissions
- RBAC propagation troubleshooting
- Credential-free VM-to-service authentication

13. Interview Questions

Q1. Why use a managed identity?
It allows an Azure resource such as a VM to authenticate to supported Azure services without storing application credentials.

Q2. What is the difference between control-plane and data-plane permissions?
Control-plane permissions manage the Azure resource itself; data-plane permissions govern operations against the data or objects inside the resource, such as Key Vault secret operations.

Q3. Why did the VM receive Key Vault Secrets User?
The VM only needs to read secrets for the planned test; it does not need permission to create or delete secrets.

Q4. Why did secret creation fail even though the administrator is Owner?
The Key Vault uses Azure RBAC for its data plane, so the administrator needs an appropriate Key Vault data role for secret creation.

Q5. Why choose Key Vault Secrets Officer instead of Administrator?
Secrets Officer is narrower and supports secret management without granting all Key Vault data operations.

Q6. Why is public network access acceptable here?
It is being used for the initial learning exercise; a production design should consider network restriction/private connectivity.

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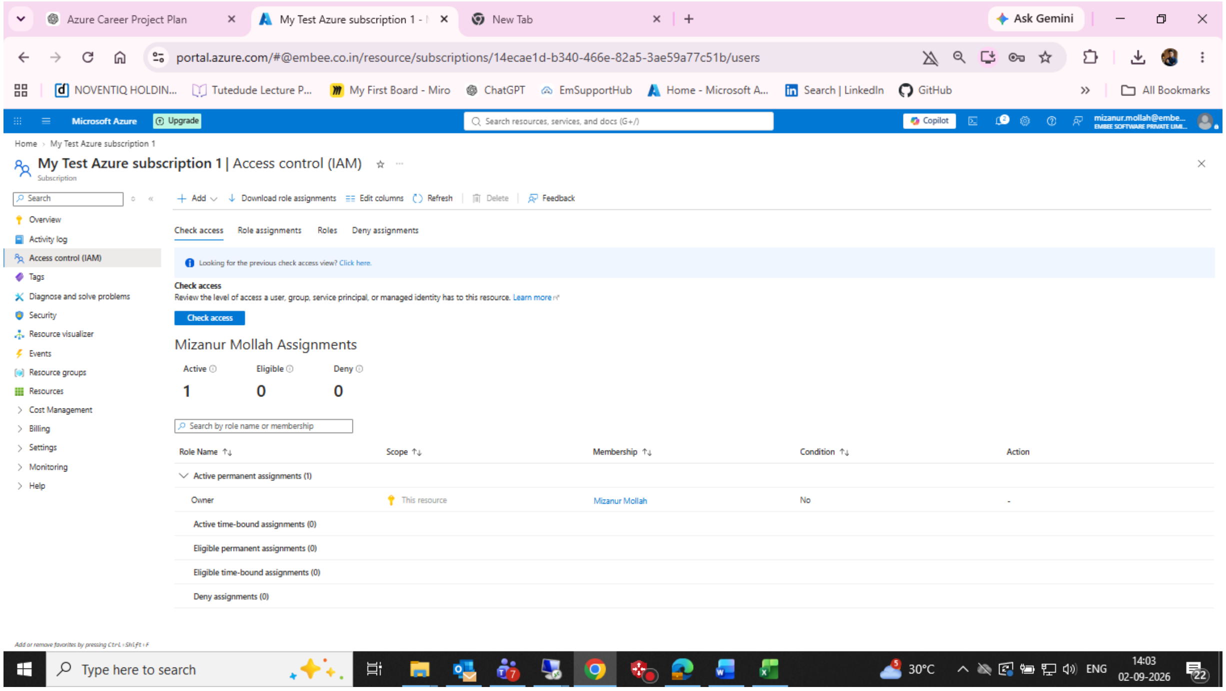
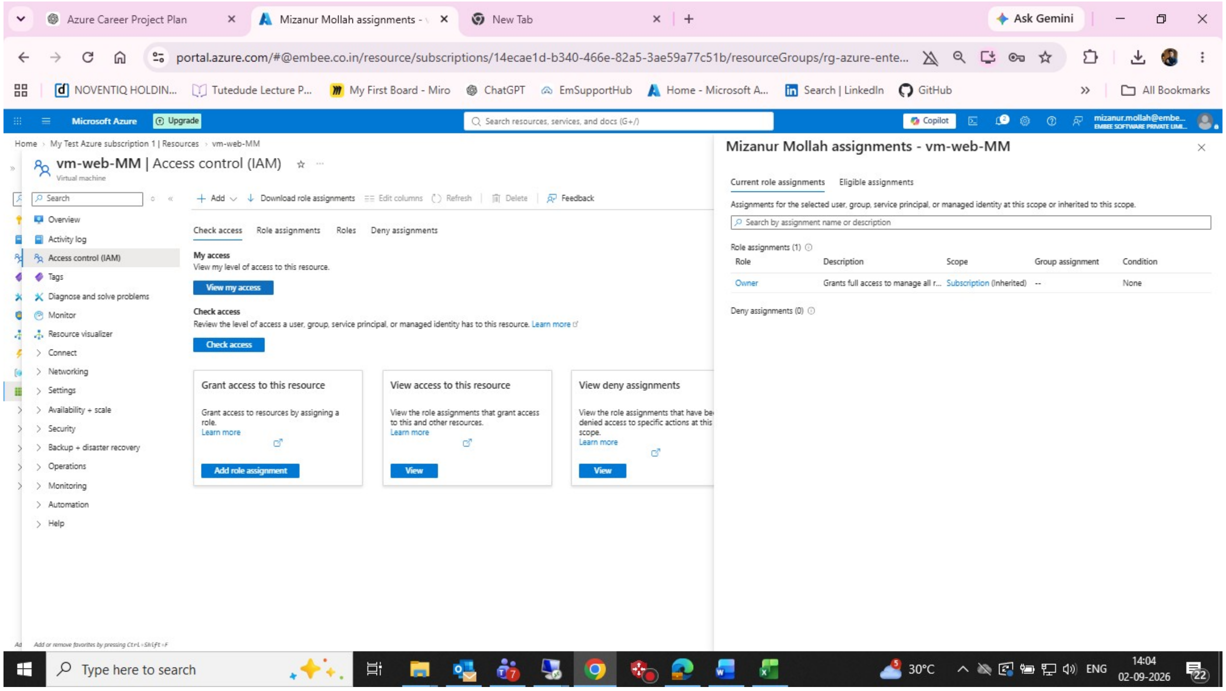
15. Screenshot Evidence

Figure 1 — Subscription IAM — Owner assignment



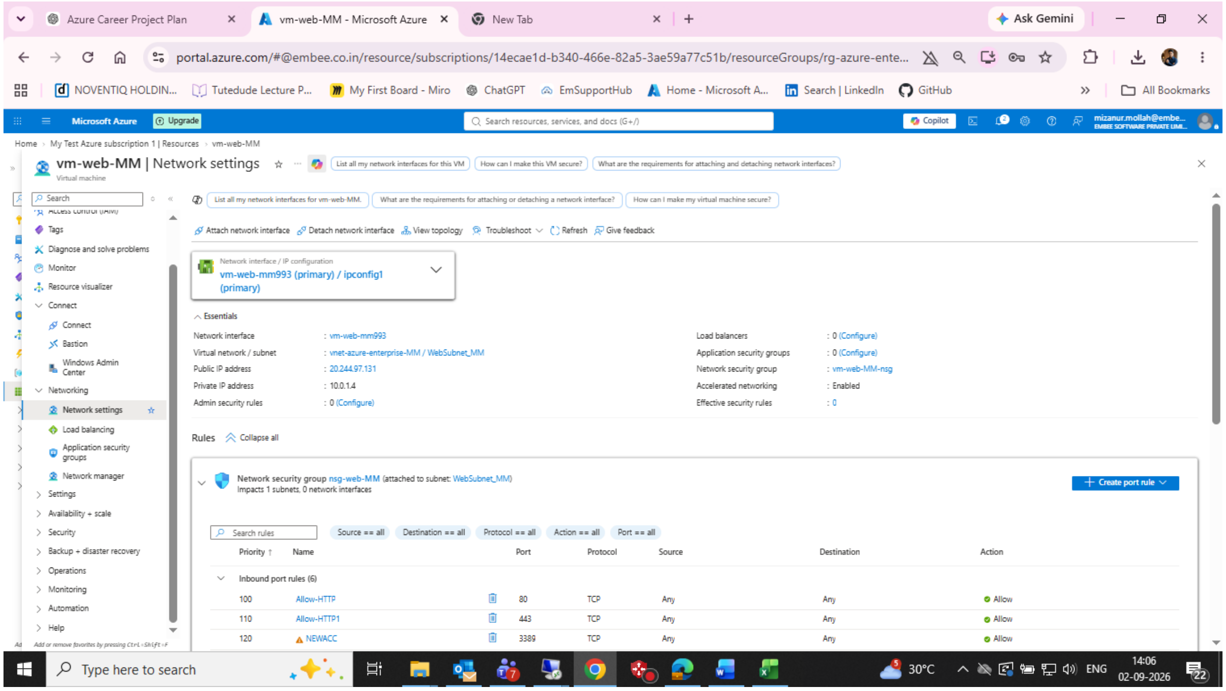
Subscription IAM — Owner assignment

Figure 2 — VM IAM — inherited Owner role



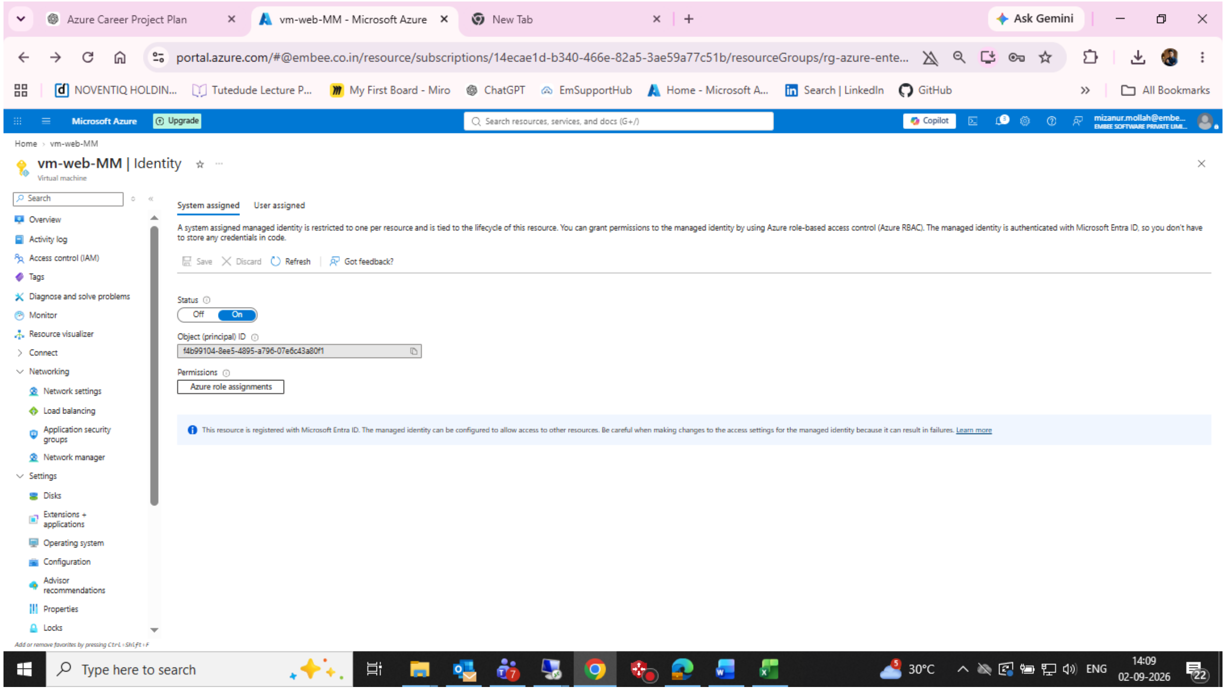
VM IAM — inherited Owner role

Figure 3 — VM Network Settings — existing NSG/network evidence



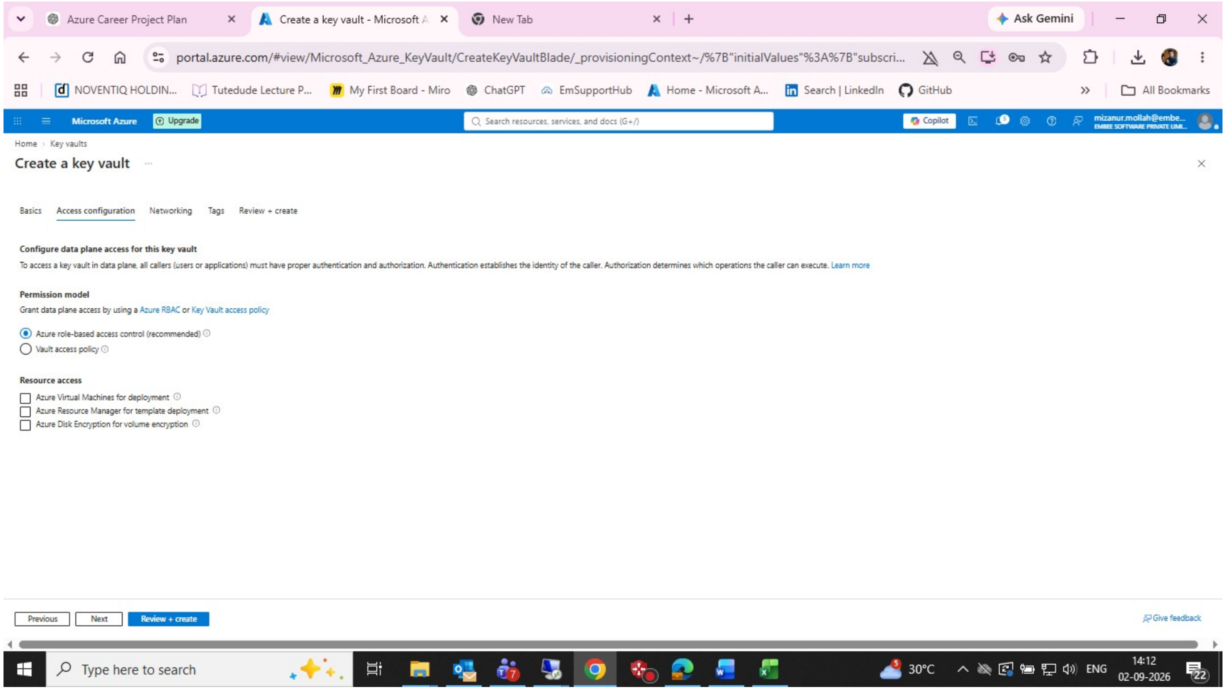
VM Network Settings — existing NSG/network evidence

Figure 4 — VM Identity — system-assigned managed identity enabled



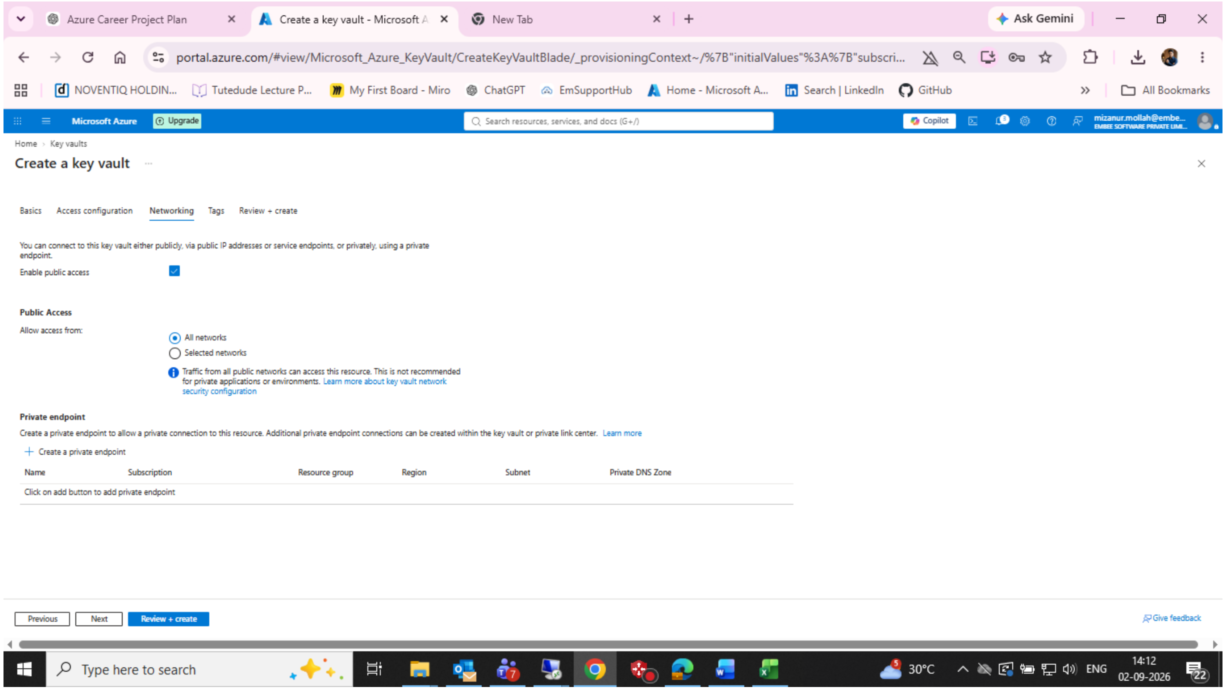
VM Identity — system-assigned managed identity enabled

Figure 5 — Key Vault Access Configuration — Azure RBAC



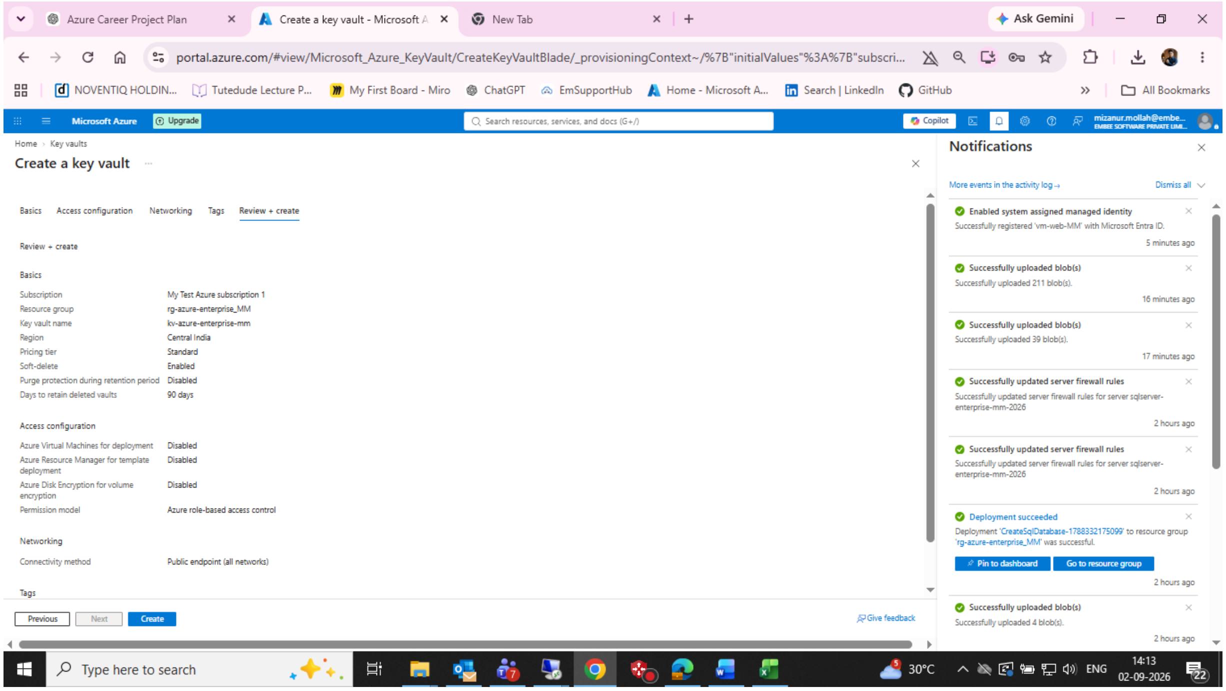
Key Vault Access Configuration — Azure RBAC

Figure 6 — Key Vault Networking — public endpoint, all networks



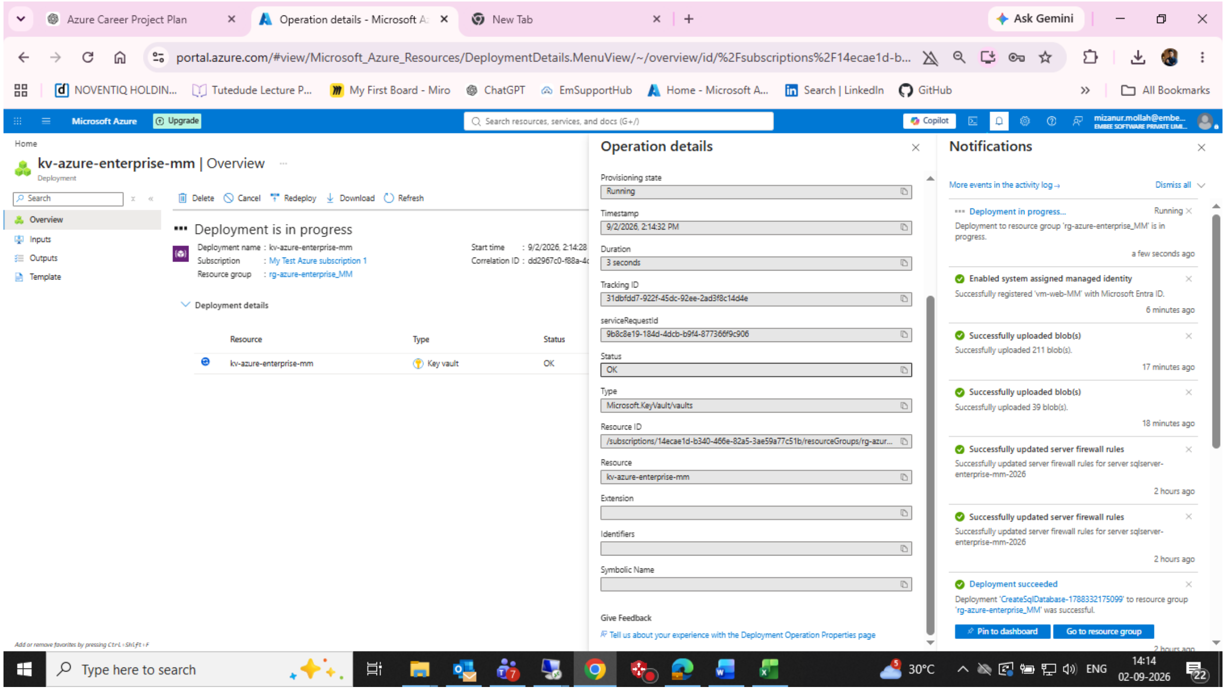
Key Vault Networking — public endpoint, all networks

Figure 7 — Key Vault Review + Create



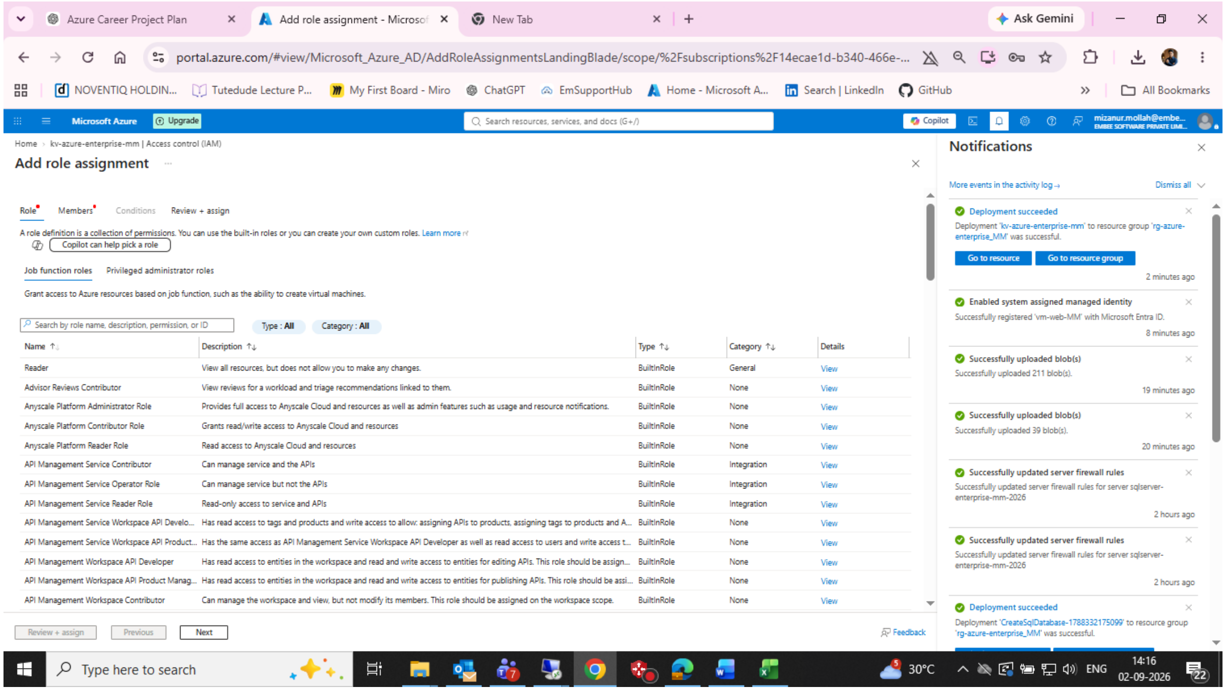
Key Vault Review + Create

Figure 8 — Key Vault deployment operation details



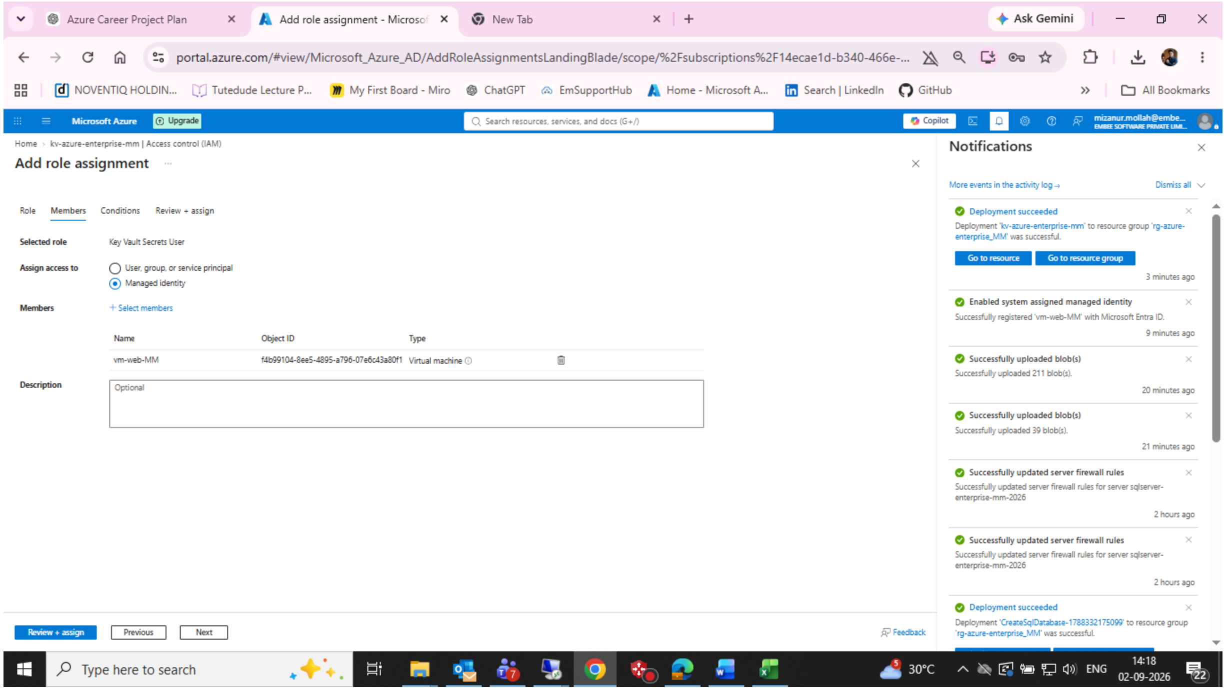
Key Vault deployment operation details

Figure 9 — Add Role Assignment — Key Vault Secrets User selected



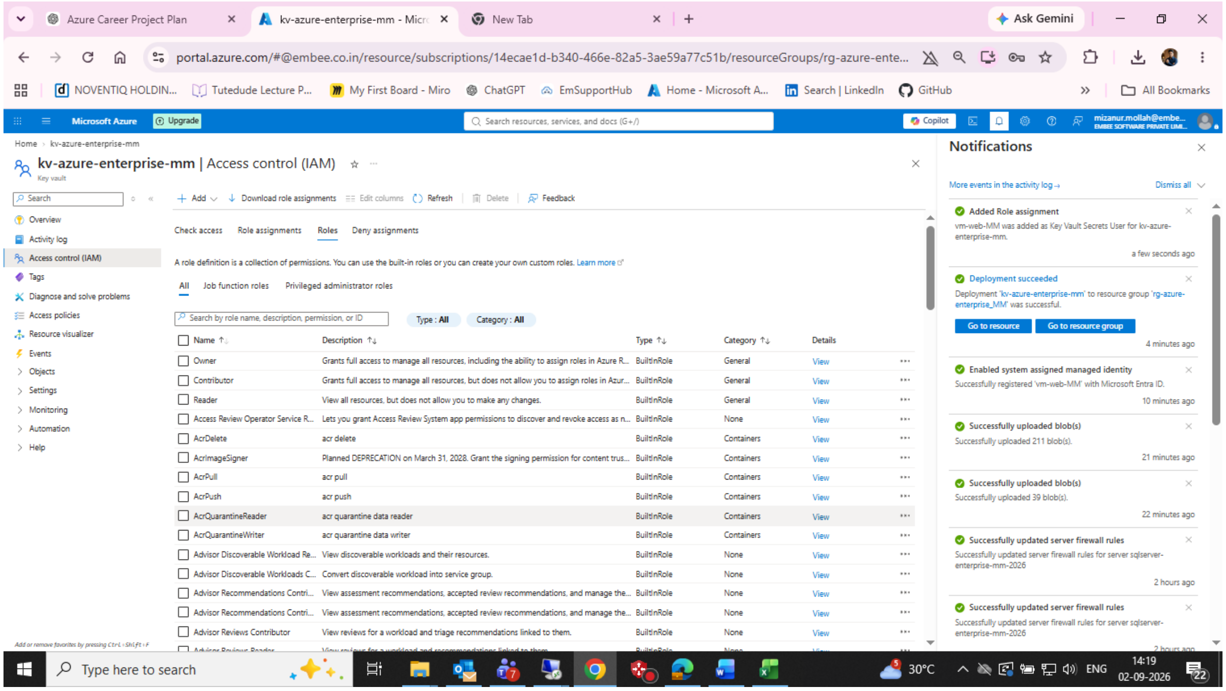
Add Role Assignment — Key Vault Secrets User selected

Figure 10 — Add Role Assignment — vm-web-MM managed identity selected



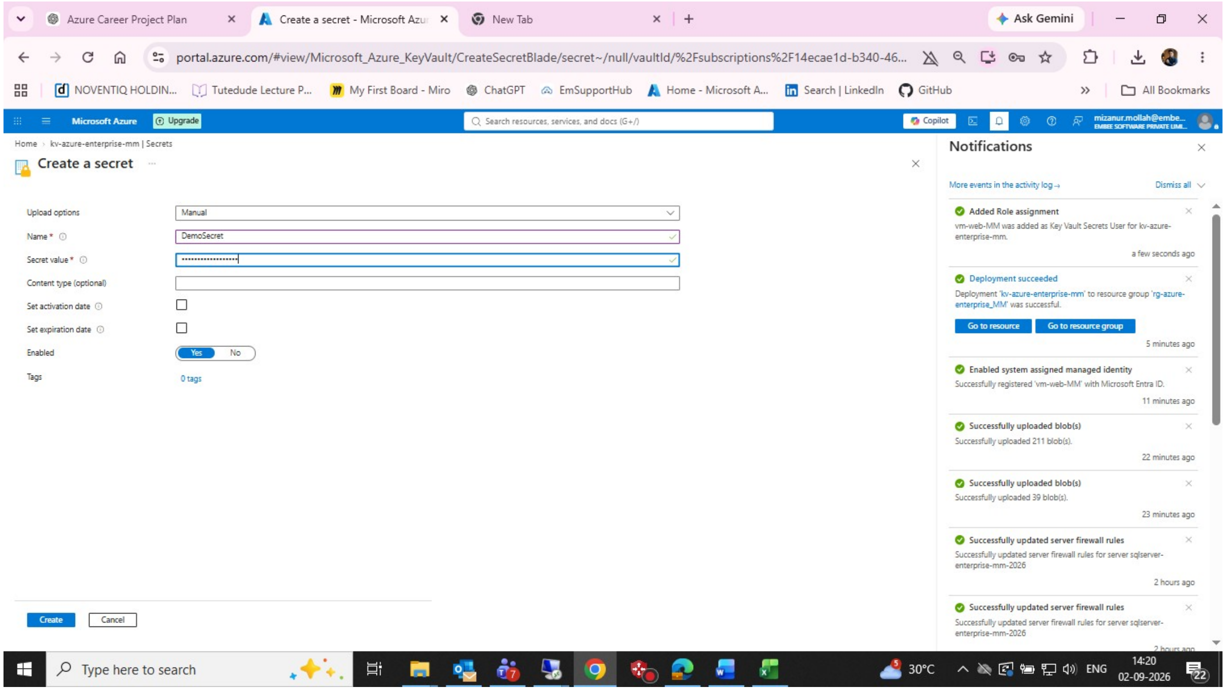
Add Role Assignment — vm-web-MM managed identity selected

Figure 11 — Key Vault RBAC role assignment success



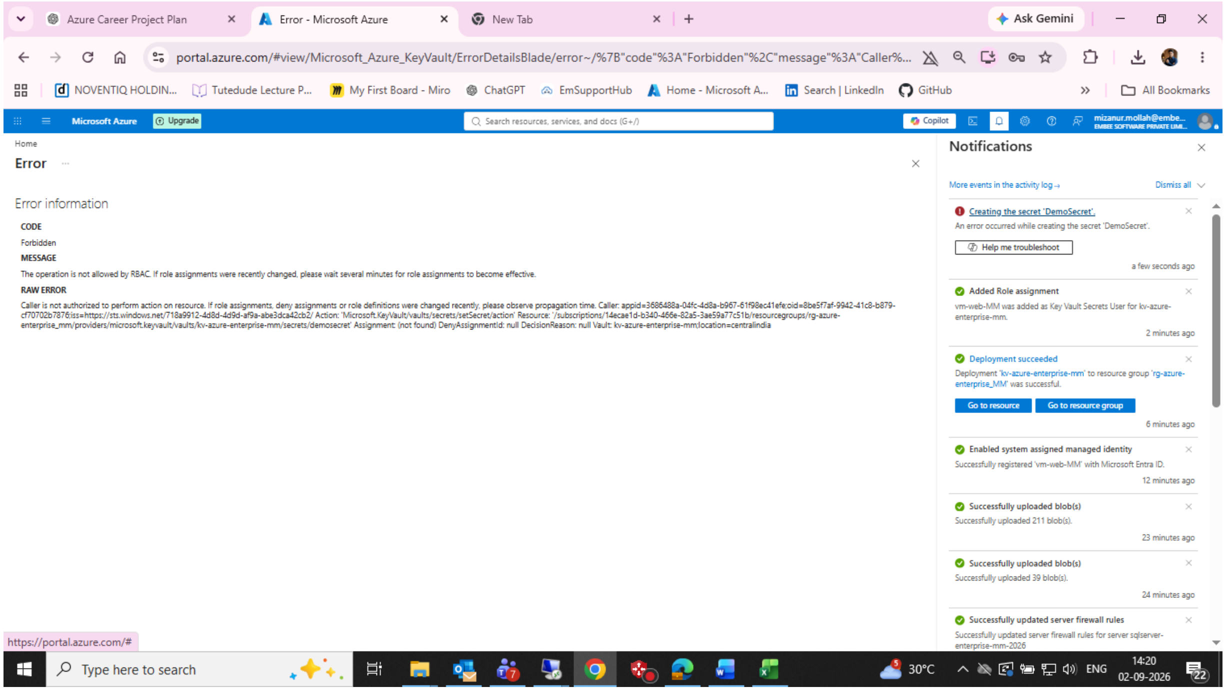
Key Vault RBAC role assignment success

Figure 12 — Create Secret — DemoSecret form



Create Secret — DemoSecret form

Figure 13 — Key Vault secret creation — Forbidden/RBAC error



Key Vault secret creation — Forbidden/RBAC error

16. GitHub Structure

- 06-identity-security/README.md
- 06-identity-security/screenshots/
- 06-identity-security/managed-identity/
- 06-identity-security/key-vault-rbac/

17. Git Commit

cd /d C:\azure-enterprise-infrastructure-lab
git add 06-identity-security
git commit -m "Document Azure identity, RBAC and Key Vault lab"
git push

18. Completion Checklist

☑️ Subscription RBAC reviewed

☑️ VM RBAC inheritance reviewed

☑️ System-assigned managed identity enabled

☑️ Key Vault created

☑️ Azure RBAC permission model selected

☑️ Key Vault Secrets User assigned to vm-web-MM

☑️ RBAC assignment success captured

☑️ DemoSecret creation attempted

☑️ RBAC Forbidden error captured

☐ Assign Key Vault Secrets Officer to administrator

☐ Wait for RBAC propagation and retry DemoSecret creation

☐ Verify DemoSecret appears in Secrets

☐ Read DemoSecret from vm-web-MM using managed identity

☐ Capture post-success screenshots

☐ Update README with final validation

☐ Sanitize personal identifiers before public GitHub upload

19. Final Status

The core identity and authorization path is configured: vm-web-MM has a system-assigned managed identity, and that identity has Key Vault Secrets User access to kv-azure-enterprise-mm. The remaining blocker is the administrator's own permission to create secrets. The documented remediation is to assign Key Vault Secrets Officer to the administrator, allow RBAC propagation, create the dummy secret, and then validate read access from the VM.